

Følner sequence

2026-10-06

1 Følner Sequences

Definition 1.1. We define a *right-Følner sequence* in Γ as a sequence $\Phi = (\Phi_N)_{N \in \mathbb{N}}$ of finite subsets of Γ satisfying

$$\lim_{N \rightarrow \infty} \frac{|(\Phi_N \cdot \gamma^{-1}) \cap \Phi_N|}{|\Phi_N|} = 1,$$

for all $\gamma \in \Gamma$.

Definition 1.2. Similarly, we define a *left-Følner sequence* in Γ as a sequence $\Phi = (\Phi_N)_{N \in \mathbb{N}}$ of finite subsets of Γ satisfying

$$\lim_{N \rightarrow \infty} \frac{|(\gamma^{-1} \cdot \Phi_N) \cap \Phi_N|}{|\Phi_N|} = 1,$$

for all $\gamma \in \Gamma$.

Definition 1.3. We call a sequence in Γ a *Følner sequence* if it is both a left and right Følner sequence.

1.1 Alternative definitions for Monoids

Definition 1.4. Let M be a countably-infinite left-cancellative monoid with discrete topology. We define a *left-Følner sequence* in M as a sequence of finite subsets $\Phi = (\Phi_N)_{N \in \mathbb{N}}$ satisfying

$$\lim_{N \rightarrow \infty} \frac{|(m \cdot \Phi_N) \cap \Phi_N|}{|\Phi_N|} = 1$$

for all $g \in M$.

Definition 1.5. Similarly, for a countably-infinite right-cancellative monoid with discrete topology M , we define a *right-Følner sequence* in M as a sequence of finite subsets $\Phi = (\Phi_N)_{N \in \mathbb{N}}$ satisfying

$$\lim_{N \rightarrow \infty} \frac{|(\Phi_N \cdot m) \cap \Phi_N|}{|\Phi_N|} = 1$$

for all $g \in M$.

1.2 Equivalent definitions using Set Differences

Equivalent definitions can be constructed by using set differences instead of intersections.

For example, the equivalent definition of a left-Følner sequence, Φ , in M requires

$$\lim_{N \rightarrow \infty} \frac{|(\Phi_N \cdot m) \triangle \Phi_N|}{|\Phi_N|} = 0,$$

to be satisfied for all $m \in M$.

This alternative definition will be useful when looking at proving some of the properties of density.

2 Tempered Følner Sequences

Definition 2.1 (Lindenstrauss, 2001, Definition 1.1). A sequence of sets $\Phi = (\Phi_N)_{N \in \mathbb{N}}$ will be said to be *tempered* if, for some $b > 0$ and all $n \in \mathbb{N}$,

$$\left| \bigcup_{1 \leq k < N} \Phi_k^{-1} \Phi_N \right| \leq b |\Phi_N|. \quad (1)$$

is referred to as the *Shulman condition*.

Proposition 2.1 (Lindenstrauss, 2001, Proposition 1.4).

1. Every Følner sequence $\Phi = (\Phi_N)_{N \in \mathbb{N}}$ has a tempered subsequence.
2. Every amenable group has a tempered Følner sequence.

Lindenstrauss, E. (2001). ‘Pointwise theorems for amenable groups’. *Inventiones mathematicae*, 146(2), pp. 259–295. <https://doi.org/10.1007/s002220100162>.